Objective kursus ini seperti berikut:

1. Pengenalan kpd Inspector berkenaan RTU - design, function, component dsbgnya.

2. Memberikan pengetahuan kepada inspector berkaitan requirement / testing untuk RTU yang perlu disemak semasa PI / FAT - ujian yg diperlukan, tujuan, acceptance criteria dan correct testing method

3. Platform kepada ES utk sanpaikan perkara - - perkara penting utk disemak semasa PI product telecontrol.

Limit Of Authority

To verify the limit of authority for each type of user.

Test Procedure & Record Example

- Open the web maintenance tool, check the User Management, and verify the task that can be done by each type of user.

User	Authority				
	Data Viewing	Control Operation	Edit Configuration	Manage User	Security Setup
Super Admin	YES ☐	YES ☐	YES ☐	YES ☐	YES ☐
Operator	YES ☐	YES ☐	NO ☐	NO ☐	NO ☐
Engineer	YES ☐	NO ☐	YES ☐	NO ☐	NO ☐
Viewer	YES ☐	NO ☐	NO ☐	NO ☐	NO ☐

##

Channel Redundancy

TENAGA NASIONAL

To confirm the capability main and redundancy for the RTU communication

Test Procedure & Record Example

No	Inspection	OK	NO
1	**IEC 101** Remove Serial Cable from Main Channel to Standby channel, then send Reset-Link. Simulate DI points to ensure the Standby Channel is online.	☐	☐
2	**IEC 104** Remove LAN/ETH Cable from Main Channel to Standby channel, send Start_DT and verify RTU communicate with Master Simulator by simulating DI point.	☐	☐
	DNP 3.0 Remove LAN/ETH Cable from Main Channel to Standby channel, send Reset-Link and verify RTU communicate with Master Simulator by simulating DI point.	☐	☐

Expected DI point reported according, with correct TID (time tag)

FAT Additional Test

Channel Redundancy

Limit Of Authority

Virtual Point & Door Alarm

Test Record Example

No	Test Parameter	IOA		
		IEC101	IEC104	DNP3.0
1	DI Fail	953 ☐	953 ☐	953 ☐
2	DO Fail	954 ☐	954 ☐	954 ☐
3	AI Fail	955 ☐	955 ☐	955 ☐
4	AO Fail	956 ☐	956 ☐	956 ☐

No	Inspection	OK	NO
1	All Virtual Point operated (TID 30)	☐	☐
2	Buzzer alarm trigger	☐	☐

Virtual Point & Door Alarm

TENAGA NASIONAL

Test Procedure

Example

Description	Procedure	Expected Result
Virtual Alarm	Simulate all virtual alarms below and observe the IOA for all virtual alarms.	The IOA shall be the same as per Master Sheet.
Door Alarm	Set Control Enable Switch (CES) in "OFF" position and close the RTU panel door	Buzzer alarm will operate /trigger

Test Record Example

Digital Output No.	Test		IOA		
	Double Command ☐				
	Open	Close			
	Single Command ☐				
	Reset		IEC101	IEC104	DNP3.0
1			☐	☐	☐
2			☐	☐	☐
3			☐	☐	☐
4			☐	☐	☐
5			☐	☐	☐
6			☐	☐	☐
7			☐	☐	☐
8			☐	☐	☐
9			☐	☐	☐
10			☐	☐	☐
11			☐	☐	☐
12			☐	☐	☐
13			☐	☐	☐
14			☐	☐	☐
15			☐	☐	☐
16			☐	☐	☐

Digital Output Module/Card

TENAGA NASIONAL

Test Record Example

No	Card Number of Total Card	Inspection	OK
1	Of	Digital Output module supplied is healthy (LED on)	
		The pilot lamp light on when "trip/close" command was sent from diagnostic tool.	

No	Inspection	OK	NO
1	The Dummy Breaker operating "Close" and "Trip" when DO is operated.	☐	☐
2	The Dummy Breaker failed to operate when Control Enable Switch is "OFF"	☐	☐

Digital Output Module/Card

TENAGA NASIONAL

Test Procedure

Example

Description	Procedure	Expected Result
Functional Test	1. Confirm each module/card number (from FAT Physical check activity). 2. Observe the LED Indicator of each Digital Output Module. 3. Simulate all Digital Output Points and observe the pilot lamp light when Trip/Close command was operated. 4. Verify the IOA is same as per Master sheet (if applicable)	1. The Digital Output Module LED should be permanently on. 2. Correct Digital Output Point is triggered once simulated.
Dummy Breaker Test	1. While CES is ON, send Trip/Close command to Dummy Breaker, and observe the response. 2. Observe the Dummy Breaker in front of the RTU Panel. 3. Repeat step 1 and 2, this time with CES OFF.	1. With CES ON, Dummy Breaker's color should change to RED (Close), and GREEN (Open/Trip) 2. The Dummy Breaker shall not operate while CES is OFF.

Analog Input Module/Card

TENAGA NASIONAL

Test Record Example

Point No	Protocol Address (IOA)	Test							OK		
		0mA	4.00mA	8.00mA	12.00mA	16.00mA	20.00mA	>20.00mA			
		0% / 0	20% / 6553	40% / 13107	60% / 19660	80% / 26214	100% / 32767	>100%			
		0.0% - 0.1%	19.9% - 20.1% / 6520-6586	39.9% - 40.1% / 13074-13140	59.9% - 60.1% / 19627-19693	79.9% - 80.1% / 26181-26247	99.9% - 100%/ 32734-32767	>100% OV	IEC101	IEC104	DNP3
1									☐	☐	☐
2									☐	☐	☐
3									☐	☐	☐
4									☐	☐	☐
5									☐	☐	☐
6									☐	☐	☐
7									☐	☐	☐
8									☐	☐	☐

** The Unipolar setting is only for the FAT purpose only (for RTU alarm circuit verification).
** During site commissioning, this setting will be changed to Live Zero.

Analog Input Module/Card

TENAGA NASIONAL

Test Procedure

Example

Description	Procedure	Expected Result
Functional Test	1. Confirm each module/card number (from FAT Physical check activity). 2. Observe the LED Indicator of each Analog Input Module. 3. Inject all Analog Input Points and observe the results. 4. Verify the IOA is same as per Master sheet (if applicable)	1. The Analog Input Module LED should be permanently on. 2. Correct Analog Input Point acquired once injected. 3. The value shall be within the accepted range.

Test Record

Example

No	Card Number of Total Card	Inspection	OK
1	Of	Analog Input module supplied is healthy (LED on)	
		The reading obtained in Analog Input Module is within range.	

Digital Input Module/Card

Test Record

Example

Digital Input No	Test		IOA		
	Double Bit ☐				
	1	2			
	Single Bit ☐				
	0	1	IEC101	IEC104	DNP3.0
1			☐	☐	☐
2			☐	☐	☐
3			☐	☐	☐
4			☐	☐	☐
5			☐	☐	☐
6			☐	☐	☐
7			☐	☐	☐
8			☐	☐	☐
9			☐	☐	☐
10			☐	☐	☐
11			☐	☐	☐
12			☐	☐	☐
13			☐	☐	☐
14			☐	☐	☐
15			☐	☐	☐
16			☐	☐	☐

Digital Input Module/Card

TENAGA NASIONAL

Test Procedure

Example

Description	Procedure	Expected Result
Functional Test	1. Confirm each module/card number (from FAT Physical check activity). 2. Observe the LED Indicator of each Digital Input module. 3. Simulate all Digital Input Points and observe the results. 4. Verify the IOA is same as per Master sheet (if 5. applicable)	1. The Digital Input Module LED should be permanently on. 2. Correct Digital Input Point is triggered once simulated.

Test Record

Example

No	Card Number of Total Card	Inspection	OK	NO
1	Of	Digital Input module supplied is healthy (LED on)	☐	☐
2	Of	Digital Input module supplied is healthy (LED on)	☐	☐

Expected Result for Start Up Process

TENAGA NASIONAL

5. Delay Acquisition (applicable for IEC101

```
--> [10:39:21] Delay Acquisition request
    Data Link: User Data  (Confirm)[PRM:1,FCV:1] [FCB:1]  Length 12 DL Addr 1
    ASDU: Type ID 106 <Delay acquisition command> Count 1 Cause of transmission  6 <Activation> Comm Addr 1
    0.000 (Seconds)
<-- [10:39:22] Ack/Fixed response
    Data Link: Ack[PRM:0,ACD:1,DFC:0]  DL Addr 1
--> [10:39:22] Class 1 request
    Data Link: Class 1 (Respond)[PRM:1,FCV:1] [FCB:0]  DL Addr 1
<-- [10:39:22] Delay Acquisition response
    Data Link: User Data[PRM:0,ACD:0,DFC:0]  Length 12 DL Addr 1
    ASDU: Type ID 106 <Delay acquisition command> Count 1 Cause of transmission  7 <Activation confirmation>
    Comm Addr 1 0.231 (Seconds)
```

6. Time sync

```
--> [10:10:01] Clock Synchronization request
    Data Link: User Data  (Confirm)[PRM:1,FCV:1] [FCB:1]  Length 17 DL Addr 1
    ASDU: Type ID 103 <Clock synchronization command> Count 1 Cause of transmission  6 <Activation> Comm Addr 1
    04/30/21 10:10:00.685 [Valid,Genuine time,Standard time]
<-- [10:10:01] Ack/Fixed response
    Data Link: Ack[PRM:0,ACD:0,DFC:0]  DL Addr 1
--> [10:10:01] Class 2 request
    Data Link: Class 2 (Respond)[PRM:1,FCV:1] [FCB:0]  DL Addr 1
<-- [10:10:01] Clock Synchronization response
    Data Link: User Data[PRM:0,ACD:0,DFC:0]  Length 17 DL Addr 1
    ASDU: Type ID 103 <Clock synchronization command> Count 1 Cause of transmission  7 <Activation confirmation>
    Comm Addr 1
    04/30/21 10:10:00.933 [Valid,Genuine time,Standard time]
```

Expected Result for Start Up Process

TENAGA NASIONAL

4. General Interrogation

```
--> [10:09:39] Interrogation request
    Data Link: User Data  (Confirm)[PRM:1,FCV:1] [FCB:0]  Length 11 DL Addr 1
    ASDU: Type ID 100 <Interrogation command> Count 1 Cause of transmission  6 <Activation> Comm Addr 1
     QOI 20 <Station interrogation>
<-- [10:09:39] Ack/Fixed response
    Data Link: Ack[PRM:0,ACD:0,DFC:0]  DL Addr 1
--> [10:09:39] Class 2 request
    Data Link: Class 2 (Respond)[PRM:1,FCV:1] [FCB:1]  DL Addr 1
<-- [10:09:39] Interrogation response
    Data Link: User Data[PRM:0,ACD:0,DFC:0]  Length 11 DL Addr 1
    ASDU: Type ID 100 <Interrogation command> Count 1 Cause of transmission  7 <Activation confirmation>
    Comm Addr 1
     QOI 20 <Station interrogation>
--> [10:09:40] Class 2 request
    Data Link: Class 2 (Respond)[PRM:1,FCV:1] [FCB:0]  DL Addr 1
<-- [10:09:40] Single Point response
    Data Link: User Data[PRM:0,ACD:0,DFC:0]  Length 12 DL Addr 1
    ASDU: Type ID 1 <Single-point> SQ Count 2 Cause of transmission  20 <Interrogated by general interrogation>
    Comm Addr 1
    DI 950 0 [Valid] DI 951 1 [Valid]
--> [10:09:41] Class 2 request
    Data Link: Class 2 (Respond)[PRM:1,FCV:1] [FCB:1]  DL Addr 1
<-- [10:09:42] Single Point response
    Data Link: User Data[PRM:0,ACD:0,DFC:0]  Length 17 DL Addr 1
    ASDU: Type ID 1 <Single-point> SQ Count 7 Cause of transmission  20 <Interrogated by general interrogation>
    Comm Addr 1
    DI 954 0 [Valid] DI 955 0 [Valid] DI 956 0 [Valid] DI 957 0 [Valid] DI 958 0 [Valid] DI 959 1 [Valid]
    DI 960 1 [Valid]
```

TID 1, 3,5 ,9 (without time)

Expected Result for Start Up Process

TENAGA NASIONAL

1. Link Status

```
--> [17:03:33] Link Status request
    Data Link: Link Status (Respond)[PRM:1,FCV:0]  DL Addr 65
<-- [17:03:33] Status of Link response
    Data Link: Status of Link[PRM:0,ACD:0,DFC:0]  DL Addr 65
--> [17:03:33] Reset Link request
    Data Link: Reset Link (Confirm)[PRM:1,FCV:0]  DL Addr 65
<-- [17:03:33] Ack/Fixed response
    Data Link: Ack[PRM:0,ACD:0,DFC:0]  DL Addr 65
```

2. Reset Link

```
--> [10:09:08] Reset Link request
    Data Link: Reset Link (Confirm)[PRM:1,FCV:0]  DL Addr 1
<-- [10:09:08] Ack/Fixed response
    Data Link: Ack[PRM:0,ACD:1,DFC:0]  DL Addr 1
```

3.End of Initialization (after power up)

```
--> [10:09:09] Class 1 request
    Data Link: Class 1 (Respond)[PRM:1,FCV:1] [FCB:1]  DL Addr 1
<-- [10:09:09] End of Initialization response
    Data Link: User Data[PRM:0,ACD:0,DFC:0]  Length 11 DL Addr 1
    ASDU: Type ID 70 <End of initialization> Count 1 Cause of transmission  4 <Initialized> Comm Addr 1
     COI 0 <Local Power Switch On> Param Unchanged
```

CPU Module/Card - Start Up Process IEC 104

TENAGA NASIONAL

IEC104

Test Record

Example

No	Inspection	OK	NO
1	Send Start_DT activation to RTU and RTU should reply Start_DT confirm.	☐	☐
2	Send Clock Sync to RTU and RTU should change it clock follow master time.	☐	☐
3	Send General Interrogations (GI) command to RTU and RTU should report all DI and AI points configured to Master Simulator	☐	☐
4	Send Broadcast Time Sync (CASDU address = 65535) to RTU and RTU should change it clock follow Master Simulator.	☐	☐

CPU Module/Card –Start Up Process IEC 101

TENAGA NASIONAL

IEC101 **Test Record** **Example**

No	Inspection	OK	NO
1	Send Reset Link Function Code to RTU and RTU should reply Link Status to Master simulator.	☐	☐
2	Send Clock Sync to RTU and RTU should change it clock follow master time.	☐	☐
3	Send General Interrogations (GI) command to RTU and RTU should report all DI and AI points configured to Master Simulator	☐	☐
4	Send Delay Acquisition to RTU and RTU should reply Master simulator with delay time.	☐	☐
5	Send Broadcast Time Sync (CASDU address = 65535) to RTU and RTU should change it clock follow Master Simulator.	☐	☐

CPU Module/Card - System Check

Test Record

Example

No	Card Number of Total Card	Inspection	OK	NO
1	Of	Processor module supplied is healthy and can be connected to maintenance tools and Master Simulator.	☐	☐
		The com port configuration setting same as per approved Drawings	☐	☐
		The I/O point setting are as per approved drawing.	☐	☐
2	Of	Processor module supplied is healthy and can be connected to maintenance tools.	☐	☐
		The com port configuration setting same as per approved Drawings	☐	☐
		The I/O point setting are as per approved drawing.	☐	☐

CPU Module/Card

TENAGA NASIONAL

Test Procedure

Example

Description	Procedure	Expected Result
Functional Test	1. Confirm each module/card number (from FAT Physical check activity). 2. Observe the LED Indicator (PWR) and (RUN). 3. Connect to the RTU using Maintenance Tools (DFETS). 4. Before start, connect ASE2000 to RTU and perform Initialization process. Ensure the RTU able to be online via (as test setup) □ IEC101 □ IEC104 □ DNP3.0.	1. The LED (PWR) indicator should be permanently on. 2. The LED (RUN) should be blinking normally. 3. Successfully connected to RTU.

Power Supply Module/Card- Power Up

Test Record

Example

No	Card Number of Total Card	Inspection	OK	NO
1	Of	Power supply module supplied is healthy (LED light shown Green).	☐	☐
2	Of	Power supply module supplied is healthy (LED light shown Green).	☐	☐

Voltage Measurement

No	Test Result	Actual Reading (Vdc)	OK	NO
1	Input Voltage		☐	☐
2	Input Voltage		☐	☐

Power Supply Module/Card

TENAGA NASIONAL

Test Procedure

Example

Description	Procedure	Expected Result
Functional Test	1. Confirm each Power Supply module/card number (from FAT Physical check activity). 2. Turn on the DC MCB, for DC Supply. 3. Turn on the switch of the Power Supply Module. 4. Observe the LED indicator of Power Supply and make sure the RTU is turned on. 5. Measure the DC Voltage of the Input.	1. The RTU shall be turned on. 2. The LED indicator should be permanently on (Green light)

MODULE FUNCTIONAL TEST

The test shall be **performed after complete power up procedure and parameter verification .** To ensure the RTU is in good working order and complies with all standards and DN TNB requirements.

The confirmation shall include RTU start up process sequence with correct TID such as:

1. **Link status**
2. **Reset Link**
3. **End of Initialization**
4. **General Interrogation**
5. **Delay acquisition.**
6. **Time sync**

All test shall be verified using Master System Protocol Simulator and record accordingly.

PARAMETER CHECK

Protocol Setting – DNP 3.0

TENAGA NASIONAL

Test Record

Example

DNP3.0

No	Test Parameter	Standard setting	OK	NO
1	Protocol No.	68. Slave_DNP	☐	☐
2	Main Channel	Ethernet A, Link 5	☐	☐
3	Standby Channel	Ethernet B Link 3	☐	☐
4	Standby Mode	Both Have Task	☐	☐
5	Port Address (if applicable)	20000	☐	☐
6	Address 1 = Source Address	1	☐	☐
7	Address 2 = Destination Address	5	☐	☐
8	Sync Time Mode	Enable	☐	☐
9	Self-Check	YES	☐	☐

PARAMETER CHECK

Protocol Setting – IEC 104

Test Record

Example

IEC104

No	Test Parameter	Standard setting	OK	NO
1	Protocol No.	114. Slave_IEC104Sec	☐	☐
2	Standby Mode	Both Have Task	☐	☐
3	Work Mode	TCP Server	☐	☐
4	Port Address	2404	☐	☐
5	Main Channel	Ethernet A, Link 2	☐	☐
6	Standby Channel	Ethernet B, Link 2	☐	☐
5	Cause of Transmission Size	2 octets	☐	☐
6	Common Address of ASDU Size	2 octets	☐	☐
7	Information Object Address Size	3 octets	☐	☐
8	Control Command Mode	CMD_BOTH	☐	☐
9	Background Scan Time (s)	0	☐	☐
10	T1 period	15 seconds	☐	☐
11	T2 period	10 seconds	☐	☐
12	T3 period	20 seconds	☐	☐
13	K	12	☐	☐
14	W	8	☐	☐

PARAMETER CHECK

Test Record

Example

Protocol Setting – IEC 101

IEC101

No	Test Parameter	Standard setting	OK	NO
1	Port	COM2 (Main), COM3 (Backup)	☐	☐
2	Protocol No.	292. Slave_IEC101Sec	☐	☐
3	Standby Mode	One working, Another	☐	☐
4	Baud rate	9600	☐	☐
5	Serial Type	RS232	☐	☐
6	Data Bits	8	☐	☐
7	Stop Bits	1	☐	☐
8	Parity	Even	☐	☐
9	R&T Under Standby	Neither Transmit nor Receive	☐	☐
10	Switching time	25 seconds	☐	☐
11	Communication Mode	Asynchronous	☐	☐
12	Sync Time Mode	Enable	☐	☐
13	Self-check Report	YES	☐	☐
14	Link Address Size	2 octets	☐	☐
15	Cause of Transmission Size	1 octet	☐	☐
16	Common Address of ASDU Size	2 octets	☐	☐
17	Information Object Address Size	2 octets	☐	☐
18	Link Transmission Procedure	Unbalanced	☐	☐
19	Alarm Time	25 seconds	☐	☐

PARAMETER CHECK

Card Setting – Digital Output

TENAGA NASIONAL

Test Record Example

Digital Output

a) Card Module

No	Test Parameter	Standard Setting		OK	NO
		Double Command	Single Command		
1	On Time	3000ms	3000ms	☐	☐
2	Reset Time	20 second	20 second	☐	☐

b) Logical Module

No	Test Parameter	Standard Setting		OK	NO
		Double Command	Single Command		
1	DO Attribute	Trip/Close	Close	☐	☐
2	Command Format	Double	Single	☐	☐
3	SBO	SBO	SBO	☐	☐
4	Priority	2	2	☐	☐

PARAMETER CHECK

Card Setting –Analog Input

Test Record Example

Analog Input

a) Card Module

No	Test Parameter	Standard Setting	OK	NO
1	AI Type	4 – 20 mA	☐	☐
2	Dead band	0	☐	☐
3	Force to Zero	0	☐	☐

b) Logical Module

No	Test Parameter	Standard Setting	OK	NO
1	Coefficient a	32767	☐	☐
2	Coefficient b	10000	☐	☐
3	Coefficient c	0	☐	☐
4	Dead band	100	☐	☐
5	Priority	2	☐	☐

Archive for AI	Disabled	Disabled	☐	
Input Signal Type for AI	Live Zero	Unipolar	☐	The unipolar setting is for FAT only (for RTU alarm circuit verification). During site commissioning, this setting will be changed to Live Zero.

PARAMETER CHECK

Card Setting – Digital Input

TENAGA NASIONAL

Test Record Example

Digital Input

a) Card Module

No	Test Parameter	Standard Setting		OK	NO
		SPI	DPI		
1	Debounce	5ms	10ms	☐	☐
2	Chatter Time = Tc	1 second	1 second	☐	☐
3	Number of Chatter = Nc	5	5	☐	☐
4	Recovery Time = TRcvt	60 seconds	60 seconds	☐	☐

b) Logical Module

No	Test Parameter	Standard Setting		OK	NO
		SPI	DPI		
1	Transmit SOE	YES	YES	☐	☐
2	Transmit COS	NO	NO	☐	☐
3	Priority	2	2	☐	☐
4	Info Code 1	(1)	(3)	☐	☐
5	Info Code 2	(30)	(31)	☐	☐
6	Event Buffer Mode	Uncovered	Uncovered	☐	☐
7	SOE Buffer Size	1000	1000	☐	☐

CPU Setting

TENAGA NASIONAL

Test Record Example

Processor Setting

11.1.2 MCU-1-A

No	Test Parameter	Standard Setting	OK	NO
1	Local GMT Time Zone	+08:00	☐	☐
2	Employ NVRAM	YES	☐	☐
3	Waiting for Next Poll	20 seconds	☐	☐
4	Clock In-Sync Period	60 minutes	☐	☐

11.1.3 MCU-4-A

No	Test Parameter	Standard Setting	OK	NO
1	Local GMT Time Zone	+08:00	☐	☐
2	Employ NVRAM	YES	☐	☐
3	Waiting for Next Poll	20 seconds	☐	☐
4	Clock In-Sync Period	60 minutes	☐	☐

PARAMETER CHECK

TENAGA NASIONAL

To confirm the RTU Parameter Setting are as approved Engineering Drawing (Tab 5).

The confirmation shall include :

1. CPU setting
2. Module setting
3. Protocol Setting

Test Procedure

Example

Description	Procedure	Expected Result
Configuration Setting	1. Open the Maintenance Tools of the RTU (DFETS). 2. Check the configuration setting for Processor, Digital Input, Digital Output, Analog Input and Analog Output. 3. Verify the setting inside the IEC101 and IEC104 or DNP3.	The setting shall be the same with the setting inside Tab 5 of Approved Engineering Drawing.

POWER UP TEST

To verify the required DC voltage to power up the RTU and AC voltage for RTU Panel

TENAGA NASIONAL

Test Procedure

Example

Description	Procedure	Expected Result
Voltage Measurement	1. On the DC MC and slowly increase the DC Voltage, until the RTU power up. 2. Record the DC and AC Voltage.	The DC and AC voltage is within the acceptable range declared.
Functionality	1. Observe the DC and Alarm Lamp. 2. Open the Panel Door and observe the Panel Light. 3. Test the AC Socket, make sure there is supply.	1. The DC Lamp should be 'ON' with Green Light, and Alarm Lamp 'ON' with Red Light. 2. Panel Light should be switched on when the Panel Door opened. 3. AC supply should be present at the socket.

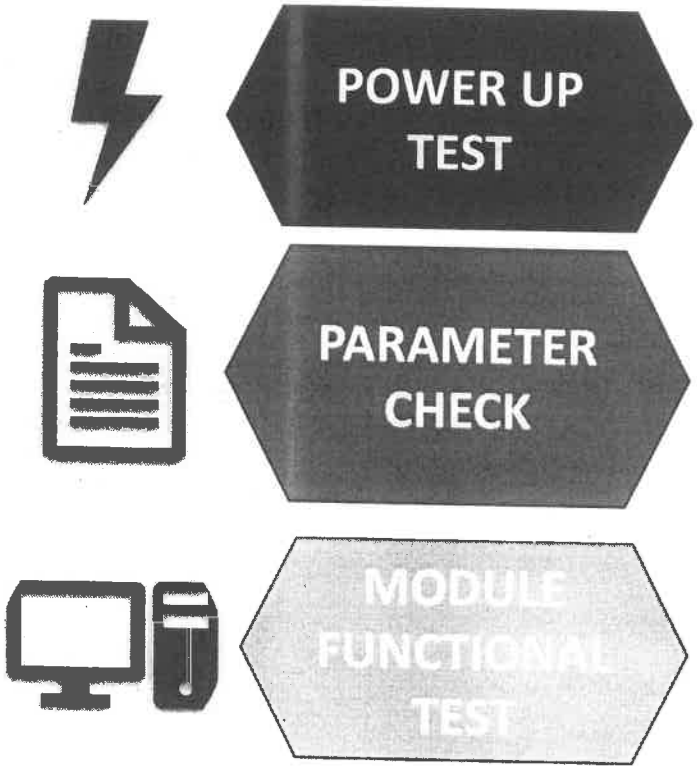
Test Record

Example

No	Test Parameter	Actual Reading	OK	NO
1	Panel Input Voltage (DC) =_____Vdc		☐	☐
2	Panel Input Voltage (AC) =_____Vac		☐	☐

No	Inspection	OK	NO
1	The DC lamp indication is 'ON' with GREEN light.	☐	☐
2	The Alarm lamp indication light is 'ON' with Red light.	☐	☐
3	The panel light is 'ON' when door panel is open.	☐	☐
4	Test 3 Pin 13A Socket 'ON'	☐	☐

FAT RTU Functional Test



Test Procedure

TENAGA NASIONAL

Example

No	Inspection Item	Procedure	Expected Result
1.0	MODULE INFORMATION	RTU Cubicle No/Panel No, RTU Rack No and Serial No. 1. Record the RTU Cubicle No, Panel No and Rack No by referring to panel label 2. Record the Serial No. by referring to the power supply module	
2.0	QUALITY	Module is free from defect e.g. deformities, corrosion, broken Inspect the module to ensure no observable physical defects e.g. sign of rust and crack	1. No observable physical defects

RTU Cards

To verify the RTU module is functioning properly and free from abnormality.

The verification shall include:

1. RTU module type
2. Quantity
3. Serial Number
4. Part Number
5. Jumper setting (Tab 2)
6. Quality (free from defect)

Example

No	Module	Info	Doc. Record Number
1	Power Supply Input	Panel Number: of	of
		Card Number: of	
		Serial Number:	
		Part Number:	
		Quality:	
		Result: ☐ Pass ☐ Fail	

RTU Panel Accessories & Component

To verify that the RTU accessories and component are fabricated, assembled and functioning as per approved Engineering Drawings. (Tab 4). The verification shall include:

1. Brand & Model
2. Quantity
3. Proper Wire marker (source & destination)
4. Quality (free from defect)

Example

No.	RTU Accessories and Components	Brand	Model	Quantity	Result (√ / X)
1	MCB DC				☐
2	MCB Auxiliary Contact				☐
3	Control Enable Switch				☐
4	DC/DC 110/24VDC Converter				☐
5	Door Switch				☐
6	Buzzer Switch				☐
7	Panel Light				☐
8	Switch Socket Outlet				☐
9	Lamp Indicator				☐
10	Lamp Indicator RTU Fault				☐
11	Terminal Block Sliding Type (Comms/DC Wetting/DO System)				☐
12	Terminal Block (I/O Plant)				☐
13	Terminal Block Sliding (DC Plant/Input/Comm)				☐
14	SPD (Comms, DC, AD)				☐
15	Modem				☐
16	Splitter				☐

Test Procedure

Example

Description	Procedure	Expected Result
Panel Information	Record the RTU Cubicle No. and RTU Panel No. by referring to the panel plat/label.	RTU Cubicle and Panel No. is the same as per Approved Engineering Drawings.
Physical Inspection	1. Observe the RTU layout installed inside the RTU Panel and compare it to the Approved Engineering Drawings.	1. The installed RTU must have same layout as in Approved Engineering Drawings.
	2. Verify the RTU Components and Accessories.	2. The RTU Components and Accessories shall match with the information inside the Approved Drawings.
	3. Check the internal wiring inside the RTU Panel.	3. Internal wiring as per RTU Cubicle Internal Wiring Diagram.
	4. Using a measuring tape, measure the height, width and depth of the RTU Panel.	4. Panel dimension is similar to the approved RTU Cubicle Drawing.
	5. Observe the physical appearance (Colour and Surface Type) of the RTU Panel.	Panel exterior, interior colour and surface type match with the approved RTU Cubicle Drawing.
Quality	1. Verify that the external and internal part of the RTU Panel is free from any defect.	1. RTU Panel shall be free from any defect, scratches and dent.
	2. Check the wiring and termination of the Panel. Make sure that the wire markers have Source and Destination.	2. The wiring should be neat, proper and tidy. It should include Source and Destination on its marker.
	3. Check the Terminal Block arrangement.	3. The Terminal Block arrangement is proper and accessible.
	4. Verify the type of cable used for Communication.	4. Shielded Twisted Pair cable is used for communication.

Panel

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To verify that the RTU panel is **fabricated, assembled and functioning as per approved Engineering Drawings**. (Tab 4).

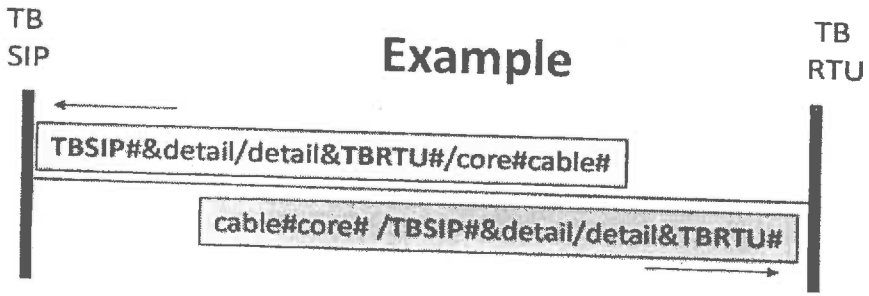
The verification shall include:

1. Panel dimension
2. Quantity
3. Serial Number
4. Proper Wire marker (source & destination)
5. Quality (free from defect)

9.1.3 Quality **Example**

No	Inspection	OK	NO
1	The Panel are free from defect eg. scratches, deformities, corrosion, broken.	☐	☐
2	The Wirings are neat, tidy, proper, and accessible.	☐	☐
3	The wire marker and ferrules are available, proper, and correct.	☐	☐
4	The wire marker has source and destination stated.	☐	☐
5	The Terminal Block arrangement is proper and tidy.	☐	☐
6	Communication cable is Shielded Twisted Pair (STP) type	☐	☐

Example



FAT RTU Physical Check

- Panel
- RTU Accessories & Components
- RTU Cards

Tester Experience and Competency

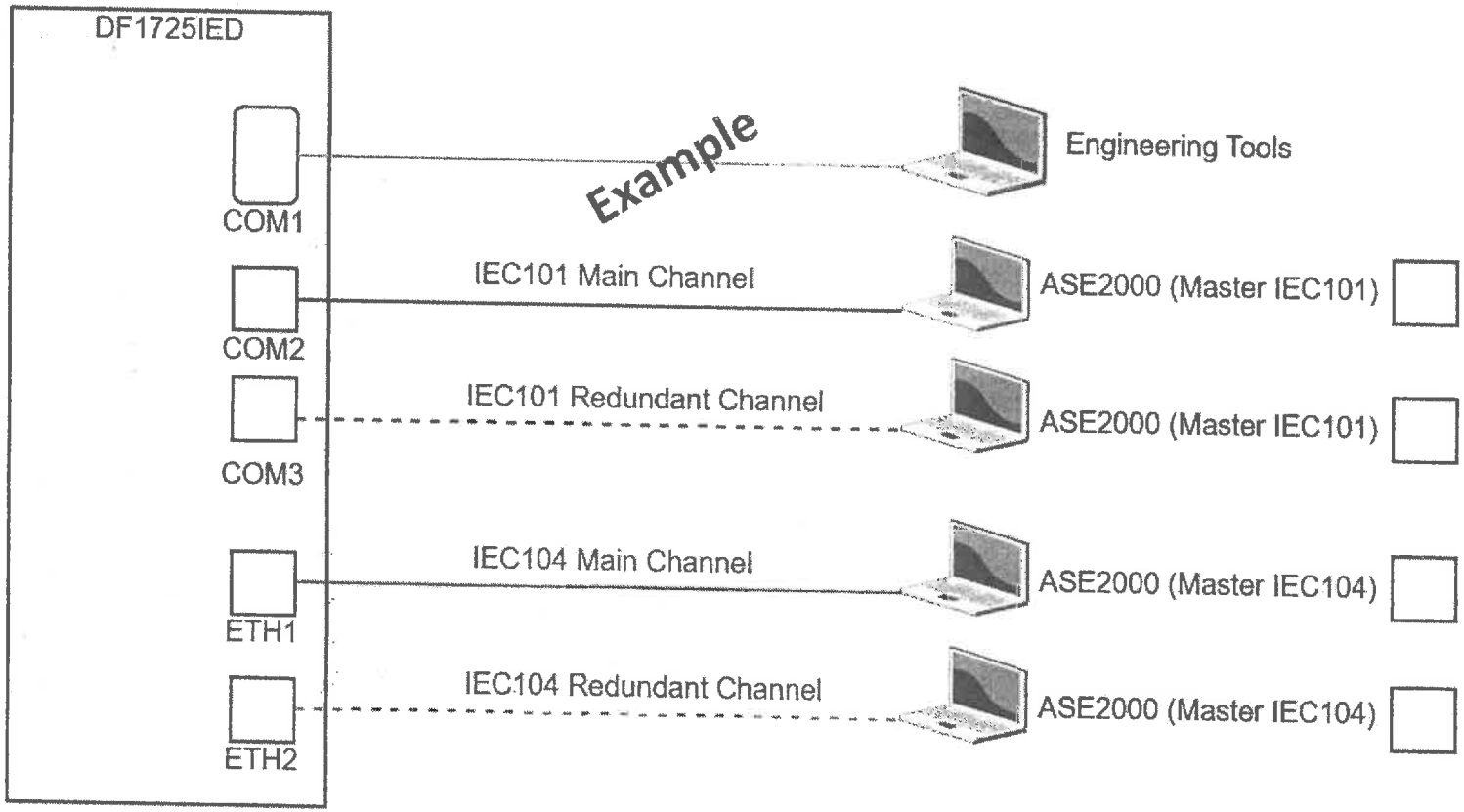
The tester is representing the contractor to conduct the test through FAT session based on approved FAT test procedure. He/she shall run the test in accordance with the test procedure and rectify any discrepancy within the test period. Minimum **5 years' experience** in the product.

Master Sheet

RTU can be configured as per RTU Information Object Addressing (IOA) format structure using RTU Configurator Software for different RTU setup. The Master Sheet shall be approved before FAT. **Drawing Tab 5.**

IO Type		IOA Address	IEC101	IEC104	DNP3.0
Digital Input	DB- Dummy Breaker	1			
	DB	2-100			
	SB- CES	101			
	SB	102-949			
	SB – Virtual Point	950-999			
Digital Output	DC- Dummy Breaker	1001			
	DC	1002-1500			
	SC	1501-2000			
Analog input	AI	2001-2099			
	AI for RE- Virtual Point	2100-2199			
Analog Output	AO	1700-1799			

FAT Test Setup



Product Declaration

	Declaration by the contractor based on approved drawing (Tab 1 & Tab 2)				Result (√ / X)
1	RTU Brand	□	RTU Model	□	
2	RTU Setup	□ Centralized	□ Sub-Master	□ Distributed	
3	RTU Protocol	□ IEC 101	□ IEC 104	□ DNP 3.0	
4	DC Input Supply	□ 110Vdc	□ 30Vdc	□ 24Vdc	
5	Comms Port	□ Single port	□ Redundancy port	□ IEDs	
6	Panel Configuration	□ Single Panel	□ Double Panel	□ Others	
7	Conformal Coated				
8	GPS				

FAT Declaration



Test Setup



Master Sheet



Tester Experience and Competency

6 RTU Accessories

TENAGA NASIONAL

To verify and record that the quantity of products supplied for the contract and as approved drawing.

No.	Description (Example)	Brand & Model	Quantity	Data Sheet	Result (√ / X)
1	Dummy Breaker				
2	DC SPD				
3	Communication SPD				
4	MCB				
5	Lamp				
6	Modem				
7	Splitter				
8	Others (based on contract)				

5 Software & Test Equipment

1. Software Record

To verify and record :
1. the RTU software version as SGP certificate
2. Master System simulator software version

No.	Description	Product Name	Brand	Version	Result (√ / X)
1	Configuration Tools				
2	Diagnostic Tools				
3	Master System Simulator				

2. Test Equipment Record

To verify the test equipment brand, model and serial number used for the Factory Acceptance Test (eg: Laptop).

No.	Description (Example)	Model	Brand	Serial Number	Result (√ / X)
1	Laptop				
2	Process Meter				

4 Certification

To verify and record the SGP certificate, panel IP certificate and test equipment calibration certificate.

No.	Certificate Title	Serial Number	Issued Date	Result (√ / X)
1	SGP			
2	Protocol Certificate IEC 101			
3	Protocol Certificate IEC 104			
4	*Protocol Certificate DNP 3			
5	*Floor Standing RTU Panel Certification IP51			
6	*Floor Standing RTU Panel Certification IP54			
7	*Wall Mounted RTU Panel Certification IP51			
8	*Wall Mounted RTU Panel Certification IP54			
9	*Measurement Tools			

*Note: * if applicable as declared in approved drawing*

3 Pre-FAT Record

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To verify and record the pre-FAT was conducted based on approved FAT test document.

The Pre-FAT record shall mention:

1. Date of testing
2. Test result
3. Verify by

Pre-FAT DATE	
Pre-FAT RESULT	☐ PASS ☐ FAIL
NOTES	

Example

2 Bill of Quantity

TENAGA NASIONAL

To verify and record that the quantity of products supplied for the contract based on approved drawing(Tab 2). The model number, part number and firmware version are as stated in SGP cert.

No.	Description (Example)	Model Number & Part Number	Firmware Version	Quantity	Result (√ / X)
1	Power Supply Unit (PSU)				
2	Central Processing Unit (CPU)				
3	Communication				
4	Digital Input (DI)				
5	Digital Output (DO)				
6	Analog Input (AI)				
7	Analog Output (AO)				
8	Others (to add based on contract)				

1 Approved Drawing

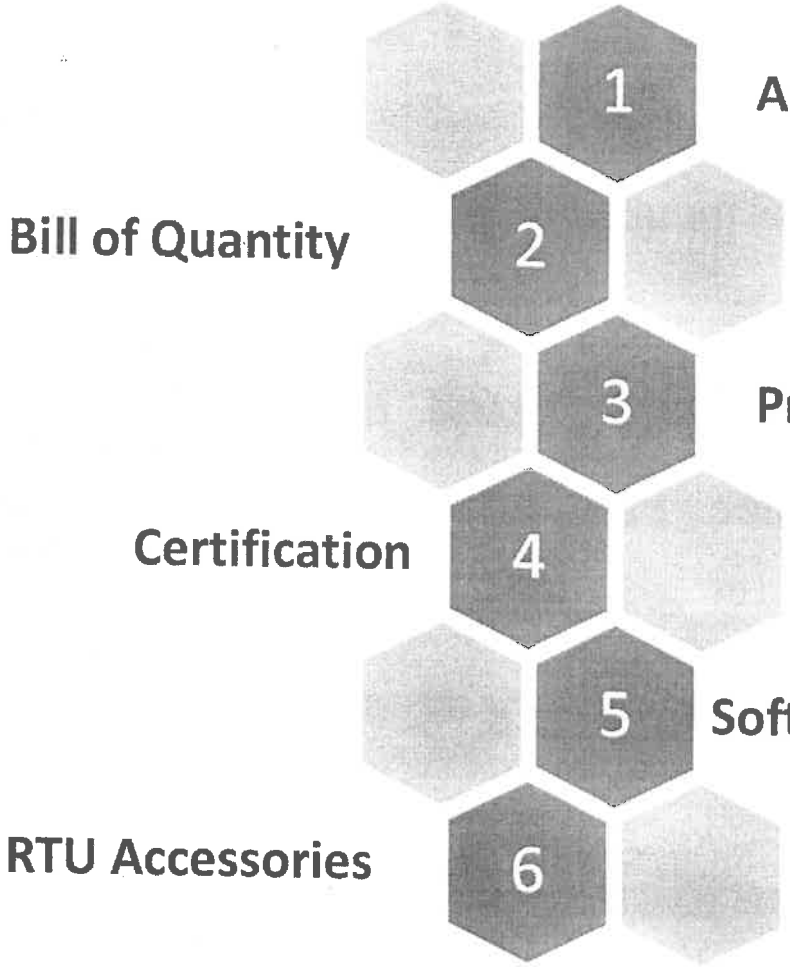
To verify that the RTU has been **fabricated and configured as approved drawing**. Details information such as Drawing No., Revision No. and Approved Date shall be recorded accordingly.

Tab No.	Drawing Title	Drawing No.	Approved Date	Rev.No	Result (√ / X)
1	Telecontrol Overview RTU Engineering Drawing				
2	RTU Configuration Engineering Drawing				
3	RTU Cubicle Internal Wiring Engineering Drawing				
4	RTU Cubicle Drawing Engineering Drawing				
5	RTU Parameter List Engineering Drawing				
6	RTU Cable Specification Engineering Drawing				

Note: Default revision number for approved SGP drawing is rev.00.

FAT Pre-Requisite & Verification

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The Objective of Factory Acceptance Test (FAT)

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The objective are to demonstrate that the RTU:

1. Has been **manufactured, assembled and configured** to meet the contract requirements, both in terms of physical hardware configuration and software configuration.
2. Is in **good working order and complies** with all the functional requirements of the contract requirements.
3. Is **free from** manufacturing and workmanship **defect**.

Factory Acceptance Test (FAT)

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START → END

PROCESS	1. PRODUCT ACCEPTANCE PROCESS (SGP)	2. TENDERING PROCESS	3. FACTORY ACCPETANCE TEST (FAT)	4. SITE ACCPETANCE TEST (SAT)	5. SITE COMMISSIONING	6. ASSET RETIREMENT
ROLES & RESPONSIBILITIES	TNB LABS TNB DN RTU VENDOR	TNB DN TNB P&SC CONTRACTOR	TNB LABS RTU VENDOR CONTRACTOR	TNB DN RTU VENDOR CONTRACTOR	TNB DN RTU VENDOR CONTRACTOR	TNB DN
REFERENCE(S) / INPUT	1. SGP application form 2. Type Test Result 3. SGP Test Guideline and procedure 4. RTU Specification 5. RTU Handbook	1. Tender document 2. BQ 3. Orange Book – *"Jaminan Dasar Kualti"*	1. Approved Drawing 2. Approved FAT Test Procedure & Record	1. Approved Drawing 2. FAT Report 3. SAT Test Procedure and Record	1. Approved Drawing 2. SAT Report 3. Commissioning Test Procedure & Record *10 -15 Years useful life*	1. Secondary Asset Replacement Policy for DN, TNB
RESULT / OUTPUT	SGP	P.O	FAT PASS FAT Report	SAT PASS SAT Report	T.O.C	Replacement Program